

Problems

See the Preface for an explanation of the icons used in this problems set. For additional assessment items for this section, go to  **WEBASSIGN** From Cengage

SECTION 2.1 Position, Velocity, and Speed

- BIO** The speed of a nerve impulse in the human body is about 100 m/s. If you accidentally stub your toe in the dark, estimate the time it takes the nerve impulse to travel to your brain.
- A particle moves according to the equation $x = 10t^2$, where x is in meters and t is in seconds. (a) Find the average velocity for the time interval from 2.00 s to 3.00 s. (b) Find the average velocity for the time interval from 2.00 to 2.10 s.
- The position of a pinewood derby car was observed at various times; the results are summarized in the following table. Find the average velocity of the car for (a) the first second, (b) the last 3 s, and (c) the entire period of observation.

t (s)	0	1.0	2.0	3.0	4.0	5.0
x (m)	0	2.3	9.2	20.7	36.8	57.5

SECTION 2.2 Instantaneous Velocity and Speed

- S** An athlete leaves one end of a pool of length L at $t = 0$ and arrives at the other end at time t_1 . She swims back and arrives at the starting position at time t_2 . If she is swimming initially in the positive x direction, determine her average velocities symbolically in (a) the first half of the swim, (b) the second half of the swim, and (c) the round trip. (d) What is her average speed for the round trip?
- A position–time graph for a particle moving along the x axis is shown in Figure P2.5. (a) Find the average velocity in the time interval $t = 1.50$ s to $t = 4.00$ s. (b) Determine the instantaneous velocity at $t = 2.00$ s by measuring the slope of the tangent line shown in the graph. (c) At what value of t is the velocity zero?

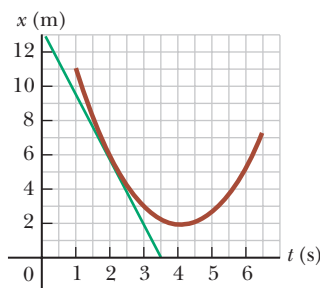


Figure P2.5

SECTION 2.3 Analysis Model: Particle Under Constant Velocity

- AMT** A car travels along a straight line at a constant speed of 60.0 mi/h for a distance d and then another distance d in the same direction at another constant speed. The average velocity for the entire trip is 30.0 mi/h. (a) What is the constant speed with which the car moved during the second distance d ? (b) **What If?** Suppose the second distance d were traveled in the opposite direction; you forgot something and had to return home at the same constant speed as found in part (a). What is the average velocity for this trip? (c) What is the average speed for this new trip?

- T** A person takes a trip, driving with a constant speed of 89.5 km/h, except for a 22.0-min rest stop. If the person's average speed is 77.8 km/h, (a) how much time is spent on the trip and (b) how far does the person travel?

SECTION 2.5 Acceleration

- A child rolls a marble on a bent track that is 100 cm long as shown in Figure P2.8. We use x to represent the position of the marble along the track. On the horizontal sections from $x = 0$ to $x = 20$ cm and from $x = 40$ cm to $x = 60$ cm, the marble rolls with constant speed. On the sloping sections, the marble's speed changes steadily. At the places where the slope changes, the marble stays on the track and does not undergo any sudden changes in speed. The child gives the marble some initial speed at $x = 0$ and $t = 0$ and then watches it roll to $x = 90$ cm, where it turns around, eventually returning to $x = 0$ with the same speed with which the child released it. Prepare graphs of x versus t , v_x versus t , and a_x versus t , vertically aligned with their time axes identical, to show the motion of the marble. You will not be able to place numbers other than zero on the horizontal axis or on the velocity or acceleration axes, but show the correct graph shapes.

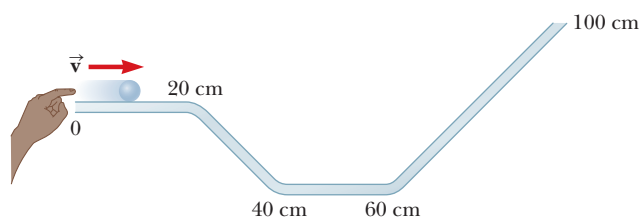


Figure P2.8

- Figure P2.9 shows a graph of v_x versus t for the motion of a motorcyclist as he starts from rest and moves along the road in a straight line. (a) Find the average acceleration for the time interval $t = 0$ to $t = 6.00$ s. (b) Estimate the time at which the acceleration has its greatest positive value and the value of the acceleration at that instant. (c) When is the acceleration zero? (d) Estimate the maximum negative value of the acceleration and the time at which it occurs.

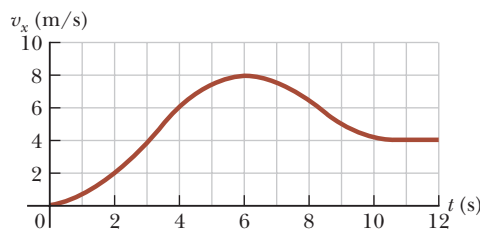


Figure P2.9

- (a) Use the data in Problem 3 to construct a smooth graph of position versus time. (b) By constructing tangents to the $x(t)$ curve, find the instantaneous velocity of the car at several instants. (c) Plot the instantaneous velocity versus time and, from this information, determine the average acceleration of the car. (d) What was the initial velocity of the car?

11. A particle starts from rest and accelerates as shown in Figure P2.11. Determine (a) the particle's speed at $t = 10.0$ s and at $t = 20.0$ s, and (b) the distance traveled in the first 20.0 s.

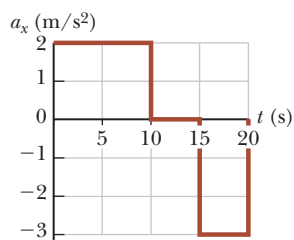
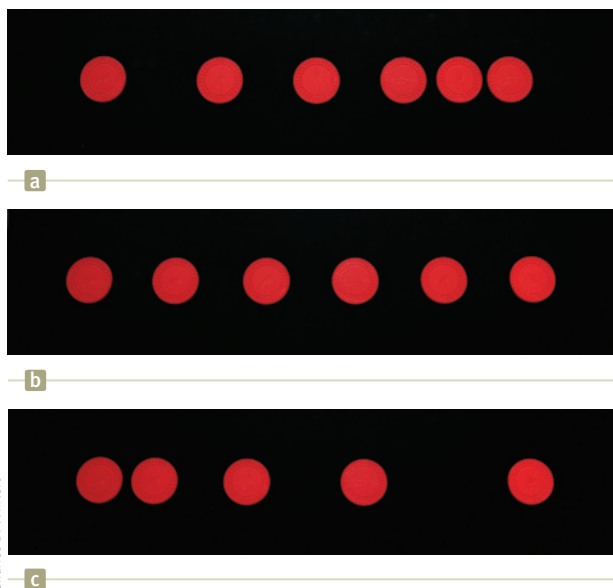


Figure P2.11

SECTION 2.6 Motion Diagrams

12. Draw motion diagrams for (a) an object moving to the right at constant speed, (b) an object moving to the right and speeding up at a constant rate, (c) an object moving to the right and slowing down at a constant rate, (d) an object moving to the left and speeding up at a constant rate, and (e) an object moving to the left and slowing down at a constant rate. (f) How would your drawings change if the changes in speed were not uniform, that is, if the speed were not changing at a constant rate?
13. Each of the strobe photographs (a), (b), and (c) in Figure P2.13 was taken of a single disk moving toward the right, which we take as the positive direction. Within each photograph the time interval between images is constant. For each photograph, prepare graphs of x versus t , v_x versus t , and a_x versus t , vertically aligned with their time axes identical, to show the motion of the disk. You will not be able to place numbers other than zero on the axes, but show the correct shapes for the graph lines.



Charles D. Winters

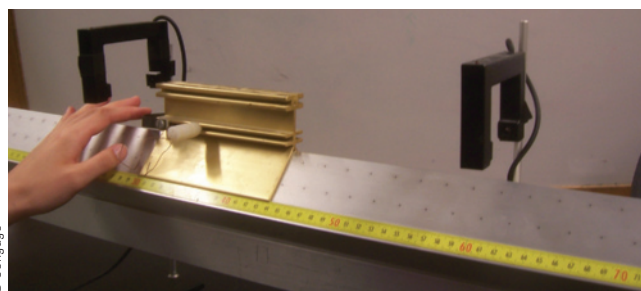
Figure P2.13

SECTION 2.7 Analysis Model: Particle Under Constant Acceleration

14. An electron in a cathode-ray tube accelerates uniformly from 2.00×10^4 m/s to 6.00×10^6 m/s over 1.50 cm. (a) In what time interval does the electron travel this 1.50 cm? (b) What is its acceleration?
15. A parcel of air moving in a straight tube with a constant acceleration of -4.00 m/s² has a velocity of 13.0 m/s at 10:05:00 a.m. (a) What is its velocity at 10:05:01 a.m.? (b) At 10:05:04 a.m.? (c) At 10:04:59 a.m.? (d) Describe the shape of a graph of velocity versus time for this parcel of air. (e) Argue for or against the following statement: "Knowing

the single value of an object's constant acceleration is like knowing a whole list of values for its velocity."

16. In Example 2.7, we investigated a jet landing on an aircraft carrier. In a later maneuver, the jet comes in for a landing on solid ground with a speed of 100 m/s, and its acceleration can have a maximum magnitude of 5.00 m/s² as it comes to rest. (a) From the instant the jet touches the runway, what is the minimum time interval needed before it can come to rest? (b) Can this jet land at a small tropical island airport where the runway is 0.800 km long? (c) Explain your answer.
17. An object moving with uniform acceleration has a velocity of 12.0 cm/s in the positive x direction when its x coordinate is 3.00 cm. If its x coordinate 2.00 s later is -5.00 cm, what is its acceleration?
18. Solve Example 2.8 by a graphical method. On the same graph, plot position versus time for the car and the trooper. From the intersection of the two curves, read the time at which the trooper overtakes the car.
19. A glider of length ℓ moves through a stationary photogate on an air track. A photogate (Fig. P2.19) is a device that measures the time interval Δt_d during which the glider blocks a beam of infrared light passing across the photogate. The ratio $v_d = \ell / \Delta t_d$ is the average velocity of the glider over this part of its motion. Suppose the glider moves with constant acceleration. (a) Argue for or against the idea that v_d is equal to the instantaneous velocity of the glider when it is halfway through the photogate in space. (b) Argue for or against the idea that v_d is equal to the instantaneous velocity of the glider when it is halfway through the photogate in time.
20. Why is the following situation impossible? Starting from rest, a charging rhinoceros moves 50.0 m in a straight line in 10.0 s. Her acceleration is constant during the entire motion, and her final speed is 8.00 m/s.
21. A glider of length 12.4 cm moves on an air track with constant acceleration (Fig P2.19). A time interval of 0.628 s elapses between the moment when its front end passes a fixed point A along the track and the moment when its back end passes this point. Next, a time interval of 1.39 s elapses between the moment when the back end of the glider passes the point A and the moment when the front end of the glider passes a second point B farther down the track. After that, an additional 0.431 s elapses until the back end of the glider passes point B. (a) Find the average speed of the glider as it passes point A. (b) Find the acceleration of the glider. (c) Explain how you can compute the acceleration without knowing the distance between points A and B.
22. In the particle under constant acceleration model, we identify the variables and parameters v_{xi} , v_{xf} , a_x , t , and



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Figure P2.19 Problems 19 and 21.

$x_f - x_i$. Of the equations in the model, Equations 2.13–2.17, the first does not involve $x_f - x_i$, the second and third do not contain a_x , the fourth omits v_{xi} , and the last leaves out t . So, to complete the set, there should be an equation *not* involving v_{xi} . Derive it from the others.

- 23. Q/C** At $t = 0$, one toy car is set rolling on a straight track with initial position 15.0 cm, initial velocity -3.50 cm/s, and constant acceleration 2.40 cm/s². At the same moment, another toy car is set rolling on an adjacent track with initial position 10.0 cm, initial velocity $+5.50$ cm/s, and constant acceleration zero. (a) At what time, if any, do the two cars have equal speeds? (b) What are their speeds at that time? (c) At what time(s), if any, do the cars pass each other? (d) What are their locations at that time? (e) Explain the difference between question (a) and question (c) as clearly as possible.

- 24. CR** You are observing the poles along the side of the road as described in the opening storyline of the chapter. You have already stopped and measured the distance between adjacent poles as 40.0 m. You are now driving again and have activated your smartphone stopwatch. You start the stopwatch at $t = 0$ as you pass pole #1. At pole #2, the stopwatch reads 10.0 s. At pole #3, the stopwatch reads 25.0 s. Your friend tells you that he was pressing the brake and slowing down the car uniformly during the entire time interval from pole #1 to pole #3. (a) What was the acceleration of the car between poles #1 and #3? (b) What was the velocity of the car at pole #1? (c) If the motion of the car continues as described, what is the number of the *last* pole passed before the car comes to rest?

SECTION 2.8 Freely Falling Objects

Note: In all problems in this section, ignore the effects of air resistance.

- 25.** *Why is the following situation impossible?* Emily challenges David to catch a \$1 bill as follows. She holds the bill vertically as shown in Figure P2.25, with the center of the bill between but not touching David's index finger and thumb. Without warning, Emily releases the bill. David catches the bill without moving his hand downward. David's reaction time is equal to the average human reaction time.



Figure P2.25

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- 26. Q/C** An attacker at the base of a castle wall 3.65 m high throws a rock straight up with speed 7.40 m/s from a height of 1.55 m above the ground. (a) Will the rock reach the top of the wall? (b) If so, what is its speed at the top? If not, what initial speed must it have to reach the top? (c) Find the change in speed of a rock thrown straight down from the top of the wall at an initial speed of 7.40 m/s and moving between the same two points. (d) Does the change in speed of the downward-moving rock agree with the magnitude of the speed change of the rock moving upward between the same elevations? (e) Explain physically why it does or does not agree.
- 27.** The height of a helicopter above the ground is given by $h = 3.00t^3$, where h is in meters and t is in seconds. At $t = 2.00$ s, the helicopter releases a small mailbag. How long after its release does the mailbag reach the ground?

- 28. T** A ball is thrown upward from the ground with an initial speed of 25 m/s; at the same instant, another ball is dropped from a building 15 m high. After how long will the balls be at the same height above the ground?

- 29. T** A student throws a set of keys vertically upward to her sorority sister, who is in a window 4.00 m above. The second student catches the keys 1.50 s later. (a) With what initial velocity were the keys thrown? (b) What was the velocity of the keys just before they were caught?

- 30. S** At time $t = 0$, a student throws a set of keys vertically upward to her sorority sister, who is in a window at distance h above. The second student catches the keys at time t . (a) With what initial velocity were the keys thrown? (b) What was the velocity of the keys just before they were caught?

- 31. CR** You have been hired by the prosecuting attorney as an expert witness in a robbery case. The defendant is accused of stealing an expensive and massive diamond ring in its box from a jewelry store. A witness to the alleged crime testified that she saw the defendant run from the store, stop next to an apartment building, and throw the box straight upward to an accomplice leaning out a fourth-floor window. When captured, the defendant did not have the stolen box with him and claimed innocence. When the witness testified in court about the defendant's throwing of the box to an accomplice, the defending attorney argued that it would be impossible to throw the box upward that high to reach the window in question. The bottom of the window is 19.0 m above the sidewalk. You have set up a demonstration in which the defendant was asked by the judge to throw a baseball horizontally as fast as he could and a radar device was used to determine that he can throw the ball at 20 m/s. (a) What testimony can you provide about the ability of the defendant to throw the box to the window in question? (b) What argument might the defense attorney make about the process used to develop your expert testimony? What might be your counter argument? Ignore any effects of air resistance on the box.

SECTION 2.9 Kinematic Equations Derived from Calculus

- 32.** A student drives a moped along a straight road as described by the velocity–time graph in Figure P2.32. Sketch this graph in the middle of a sheet of graph paper. (a) Directly above your graph, sketch a graph of the position versus time, aligning the time coordinates of the two graphs. (b) Sketch a graph of the acceleration versus time directly below the velocity–time graph, again aligning the time coordinates. On each graph, show the numerical values of x and a_x for all points of inflection. (c) What is the acceleration at $t = 6.00$ s? (d) Find the position (relative to the starting point) at $t = 6.00$ s. (e) What is the moped's final position at $t = 9.00$ s?

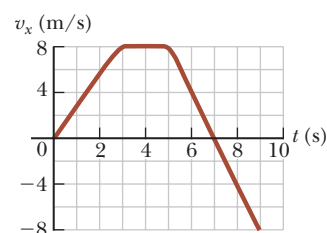


Figure P2.32

- 33. S** Automotive engineers refer to the time rate of change of acceleration as the “jerk.” Assume an object moves in one dimension such that its jerk J is constant. (a) Determine expressions for its acceleration $a_x(t)$, velocity $v_x(t)$, and position $x(t)$, given that its initial acceleration, velocity, and position are a_{xi} , v_{xi} , and x_i , respectively. (b) Show that $a_x^2 = a_{xi}^2 + 2J(v_x - v_{xi})$.

ADDITIONAL PROBLEMS

- 34.** In Figure 2.11b, the area under the velocity–time graph and between the vertical axis and time t (vertical dashed line) represents the displacement. As shown, this area consists of a rectangle and a triangle. (a) Compute their areas. (b) Explain how the sum of the two areas compares with the expression on the right-hand side of Equation 2.16.
- 35.** The froghopper *Philaeus spumarius* is supposedly the best jumper in the animal kingdom. To start a jump, this insect can accelerate at 4.00 km/s^2 over a distance of 2.00 mm as it straightens its specially adapted “jumping legs.” Assume the acceleration is constant. (a) Find the upward velocity with which the insect takes off. (b) In what time interval does it reach this velocity? (c) How high would the insect jump if air resistance were negligible? The actual height it reaches is about 70 cm , so air resistance must be a noticeable force on the leaping froghopper.
- 36.** A woman is reported to have fallen 144 ft from the 17th floor of a building, landing on a metal ventilator box that she crushed to a depth of 18.0 in. She suffered only minor injuries. Ignoring air resistance, calculate (a) the speed of the woman just before she collided with the ventilator and (b) her average acceleration while in contact with the box. (c) Modeling her acceleration as constant, calculate the time interval it took to crush the box.
- 37.** At $t = 0$, one athlete in a race running on a long, straight track with a constant speed v_1 is a distance d_1 behind a second athlete running with a constant speed v_2 . (a) Under what circumstances is the first athlete able to overtake the second athlete? (b) Find the time t at which the first athlete overtakes the second athlete, in terms of d_1 , v_1 , and v_2 . (c) At what minimum distance d_2 from the leading athlete must the finish line be located so that the trailing athlete can at least tie for first place? Express d_2 in terms of d_1 , v_1 , and v_2 by using the result of part (b).
- 38.** Why is the following situation impossible? A freight train is lumbering along at a constant speed of 16.0 m/s . Behind the freight train on the same track is a passenger train traveling in the same direction at 40.0 m/s . When the front of the passenger train is 58.5 m from the back of the freight train, the engineer on the passenger train recognizes the danger and hits the brakes of his train, causing the train to move with acceleration -3.00 m/s^2 . Because of the engineer’s action, the trains do not collide.
- 39.** Hannah tests her new sports car by racing with Sam, an experienced racer. Both start from rest, but Hannah leaves the starting line 1.00 s after Sam does. Sam moves with a constant acceleration of 3.50 m/s^2 , while Hannah maintains an acceleration of 4.90 m/s^2 . Find (a) the time at which Hannah overtakes Sam, (b) the distance she travels before she catches him, and (c) the speeds of both cars at the instant Hannah overtakes Sam.
- 40.** Two objects, A and B, are connected by hinges to a rigid rod that has a length L . The objects slide along perpendicular guide rails as shown in Figure P2.40. Assume object A slides to the left with a constant speed v . (a) Find the velocity v_B of object B as a function of the angle θ . (b) Describe v_B

relative to v . Is v_B always smaller than v , larger than v , or the same as v , or does it have some other relationship?

- 41.** Lisa rushes down onto a subway platform to find her train already departing. She stops and watches the cars go by. Each car is 8.60 m long. The first moves past her in 1.50 s and the second in 1.10 s . Find the constant acceleration of the train.

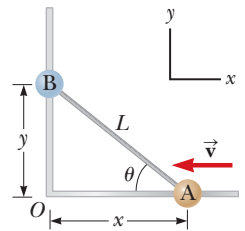


Figure P2.40

CHALLENGE PROBLEMS

- 42.** Two thin rods are fastened to the inside of a circular ring as shown in Figure P2.42. One rod of length D is vertical, and the other of length L makes an angle θ with the horizontal. The two rods and the ring lie in a vertical plane. Two small beads are free to slide without friction along the rods. (a) If the two beads are released from rest simultaneously from the positions shown, use your intuition and guess which bead reaches the bottom first. (b) Find an expression for the time interval required for the red bead to fall from point A to point C in terms of g and D . (c) Find an expression for the time interval required for the blue bead to slide from point B to point C in terms of g , L , and θ . (d) Show that the two time intervals found in parts (b) and (c) are equal. *Hint:* What is the angle between the chords of the circle A–B and B–C? (e) Do these results surprise you? Was your intuitive guess in part (a) correct? This problem was inspired by an article by Thomas B. Greenslade, Jr., “Galileo’s Paradox,” *Phys. Teach.* **46**, 294 (May 2008).
- 43.** In a women’s 100-m race, accelerating uniformly, Laura takes 2.00 s and Healan 3.00 s to attain their maximum speeds, which they each maintain for the rest of the race. They cross the finish line simultaneously, both setting a world record of 10.4 s . (a) What is the acceleration of each sprinter? (b) What are their respective maximum speeds? (c) Which sprinter is ahead at the 6.00-s mark, and by how much? (d) What is the maximum distance by which Healan is behind Laura, and at what time does that occur?
- 44.** **Review.** You are sitting in your car at rest at a traffic light with a bicyclist at rest next to you in the adjoining bicycle lane. As soon as the traffic light turns green, your car speeds up from rest to 50.0 mi/h with constant acceleration 9.00 mi/h/s and thereafter moves with a constant speed of 50.0 mi/h . At the same time, the cyclist speeds up from rest to 20.0 mi/h with constant acceleration 13.0 mi/h/s and thereafter moves with a constant speed of 20.0 mi/h . (a) For what time interval after the light turned green is the bicycle ahead of your car? (b) What is the maximum distance by which the bicycle leads your car during this time interval?

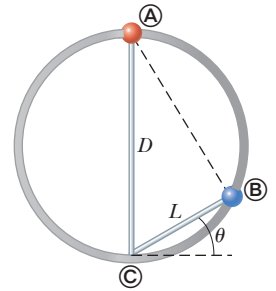


Figure P2.42